

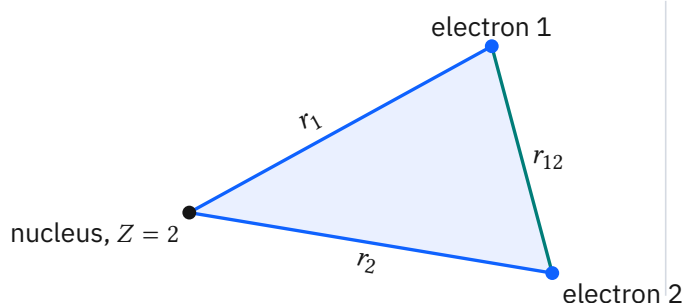
Helium

Toward an exact analytic wavefunction

Research program with a guiding literature supplement

Determine the ground state, not another approximation to it.

Construct a globally valid expression for the wavefunction and energy, with every connection parameter fixed by the physical problem. No fitted exact energy. No unknown angular function. No Schrödinger equation hidden behind a new function name.



Coordinate geometry only; not an orbit or a computed wavefunction.

THREE DISTANCES

$$\psi(r_1, r_2, r_{12})$$

ONE PHYSICAL MINIMUM

$$H_2\psi = E\psi$$

Working hypothesis. Retain the singular and global compatibility structure long enough to derive an exact elimination. The enlarged formulation earns its place only when it removes an unknown.

PRIMARY ENDPOINT

Nonrelativistic, fixed-nucleus helium; an exact spatial ground state, its energy, and a proof of minimum selection.

PROGRAM STATUS

Kickoff charter, version 1.0. No helium-specific analytic closure is claimed in this document.

12 September 2026

Research program / not a results paper

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Contents & reading guide

One physical target. One standard of exactness.

The program states the objective, proof obligations, and launch protocol. The supplement identifies the closest established methods, the global determination they leave unresolved, and the literature still requiring inspection.

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HOW THE SUPPLEMENT GUIDES THE NEXT EXPERIMENTS

01 Global connection

Fix the Fock expansion's remaining data from global conditions. Compare the proposed elimination with the existing recurrence and resummation work.
Supplement §§3, 10; records R03-R05 and R17-R18.

02 Generator resummation

Start beyond Free Complement's established construction-and-verification machinery. Identify the coefficient-selection step the new law would remove.
Supplement §§4, 10; records R07-R11.

03 Exact pair-kernel completion

Use known two-body Coulomb information without leaving the coupled three-body amplitude or physical-root selection unresolved.
Supplement §§6, 10; record R25.

Read the revised research target on page 16 before launching a candidate. The audit guides the charter; it does not replace its exactness standard. Original section and reference identifiers remain local to each part.

01

The physical target

A smaller question than universal DFT, with the same standard of genuine elimination.

Mission. Derive an exact analytic representation of the nonrelativistic helium ground state and determine its energy and all connection data from the Hamiltonian. The density is an output, not an exact input borrowed from a reference calculation.

In atomic units, the target operator is

$$H_2 = -\frac{1}{2}(\nabla_1^2 + \nabla_2^2) - \frac{2}{r_1} - \frac{2}{r_2} + \frac{1}{r_{12}}. \quad (1)$$

The nucleus is an infinitely heavy point charge fixed at the origin. The task is the ground 1^1S state. Its normalized spin factor is the singlet; the spatial factor is symmetric, rotationally invariant, and can be chosen real and positive. The standard three-distance representation is described in the helium literature. [4][5]

Domain, normalization, and the selected state

The internal domain is

$$\mathcal{D} = \{r_1, r_2 \geq 0, |r_1 - r_2| \leq r_{12} \leq r_1 + r_2\}.$$

The spatial function must satisfy

$$\psi(r_1, r_2, r_{12}) = \psi(r_2, r_1, r_{12}), \quad 8\pi^2 \int_{\mathcal{D}} r_1 r_2 r_{12} |\psi|^2 dr_1 dr_2 dr_{12} = 1.$$

The measure follows by integrating the overall orientation of the electron–nucleus triangle. Collinear faces are coordinate boundaries, not physical hard walls: do not impose artificial zeros there.

A completed solution belongs to the physical Coulomb Hamiltonian domain, is square-integrable with finite kinetic energy, has the correct coalescence behavior, and decays in every escape channel. Its energy is below the helium-ion-plus-electron threshold, -2 hartree. Local formulas must join into one physical state; boundary data may not remain arbitrary.

Inputs, outputs, and deferred extensions

Inputs	Required outputs
$Z = 2$, electron masses and charges, and the stated units and boundary problem.	An explicit ψ , E_0 , normalization, every connection coefficient or branch parameter, and a proof that this is the minimum.

Finite nuclear mass, relativistic and radiative corrections, excited states, magnetic fields, and a universal density functional are deferred. They are not silently incorporated into the claim. Varying Z or the electron–electron coupling is permitted as a derivation or stress test, but exact helium at the specified Hamiltonian remains the endpoint.

Scope principle. Solving helium need not wait for a universal density functional. An exact helium result would not, by itself, establish one.

What will count as closure

Exact can be non-elementary. It cannot be circular.

Primary deliverable. A fully determined construction for ψ and E that satisfies $H_2\psi = E\psi$ on the physical domain, meets all boundary conditions, and selects the ground state. Every remaining constant and branch is fixed by explicit, independently evaluable rules.

Admissible forms of an exact answer

A finite expression in known functions would qualify if proved. So could a genuinely evaluated integral transform, a non-elementary special-function representation, or a convergent generating law with explicitly determined coefficients and globally controlled continuation.

A scalar quantization equation is acceptable only when its coefficients are constructible from the stated inputs and the physical root is selected independently. A recurrence is acceptable only when it determines its data, has a proved domain of validity, and does not ask for a fresh unrestricted angular solution at each order.

A formal spectral expansion can be exact and still fall below this program’s research bar. The proposed construction must demonstrate a real reduction in the information or computation needed to determine the state beyond standard systematic diagonalization. Count coefficient generation, required precision, branch selection, and verification as part of the cost.

What does not meet the endpoint

Attractive substitute	Why the job remains open
A highly accurate trial energy	Does not establish the complete wavefunction or a zero residual.
A finite-grid or basis certificate	Certifies its discretized problem unless the continuum error is separately controlled.
An exact reformulation	Fails if its unknown kernel, density, conditional state, or inverse contains the original solve.
An implicit “new function”	Fails if its definition is the same three-variable Schrödinger equation, with no new determining mechanism.
A fitted energy or connection parameter	Uses part of the desired answer as an input. Reference values belong in validation only.

A fallback milestone is not a renamed victory

A sequence of certified numerical approximations would be an important intermediate result. A scoped impossibility theorem may retire an ansatz. Neither is automatically the requested exact closed representation. The final claim must state exactly which level has been reached.

Dependency audit. List every unknown scalar, function, kernel, spectrum, measure, and boundary value, with its determining rule and evaluation cost. An unexplained entry blocks a closure claim.

03

The frontier to surpass

The challenge is not another accurate helium calculation.

Known achievements are the baseline

High precision. Schwartz's 2006 explicitly correlated expansion used 24,099 terms and reported more than forty digits of the idealized helium energy. Its value begins [3]

$$E_0 = -2.9037243770341195983 \dots \text{ hartree.}$$

This is a validation reference, not a constant allowed inside the proposed construction. The project does not need to win a digit-count contest to make an analytic advance.

Compact analytical approximations. Few-parameter correlated expressions and cusp-aware forms already achieve useful accuracy across helium-like ions. A new short ansatz needs comparison with that literature, not only with an uncorrelated product. [4]

Singularity-aware expansions. Kato's coalescence conditions, Fock logarithms, and exponential forms incorporating them are established. Modern Fock-aware calculations still determine coefficients through a generalized eigenproblem. The intended contribution is to eliminate or fundamentally simplify that remaining determination. [1][2][5]

Lower as well as upper bounds. Published few-electron calculations already provide tight lower bounds comparable with variational upper bounds. Our certificate must be evaluated against that work, not presented as novel merely because it is two-sided. [6]

Recent numerical and interpolation work. An August 2026 preprint reports high-precision Lagrange-mesh energies and compact energy interpolations for helium-like ions. An interpolation of energies is not an exact spatial wavefunction. [7]

The research gap we intend to attack

Determine the global connection between known local Coulomb structure and physical decay, so that no independent angular function, fitted energy, or unrestricted coefficient problem remains.

What we carry forward from the DFT program

The inherited rules are methodological: retain compatibility until it is genuinely eliminated; test the minimum rather than stationarity alone; distinguish an energy identity from a method for determining its inputs; and treat apparent exactness on special models as a control, not evidence of helium closure.

Prior shell-density calculations and lattice constructions are not helium proofs. Their techniques must be rederived with the nuclear cusps, the correct measure, and the unbounded Coulomb domain. The unfinished DFT idea backlog is not reactivated by this charter.

Novelty gate. Before substantial development, compare the candidate with its closest primary antecedent. Published claims of an "exact" or "analytic" helium solution must be checked against the same Hamiltonian, domain, and no-hidden-input contract. A title is not the acceptance test.

04

The first analytical attack

Build in the singular structure; solve its global completion.

Constraints that must survive every representation

Away from the simultaneous triple collision, the spherical-average cusp conditions are [1][2]

$$\left. \frac{\partial \bar{\psi}}{\partial r_i} \right|_{r_i=0} = -2 \bar{\psi}(0), \quad \left. \frac{\partial \bar{\psi}}{\partial r_{12}} \right|_{r_{12}=0} = \frac{1}{2} \bar{\psi}(0).$$

The electron–nucleus average holds with the other electron fixed; the electron–electron average uses relative separation at fixed pair center. These are not arbitrary unaveraged derivatives at the triple collision.

Near that triple collision, set $R = \sqrt{r_1^2 + r_2^2}$ and let Ω denote the two internal angular variables. The established local form is [2][5]

$$\psi(R, \Omega) = \sum_{n \geq 0} R^n \sum_{p=0}^{\lfloor n/2 \rfloor} [\ln(R/a_0)]^p f_{np}(\Omega).$$

Here a_0 is the unit of length. A local expansion is not a proof of global determination. Track all homogeneous angular freedoms and show how they are fixed by global conditions; do not insert them from a reference state.

An immediate exact falsification control

The familiar cusp-correct seed [2]

$$\psi_{\text{seed}} = \exp[-2(r_1 + r_2) + r_{12}/2]$$

has, by direct differentiation,

$$\frac{H_2 \psi_{\text{seed}}}{\psi_{\text{seed}}} = -\frac{17}{4} + \frac{(r_1 + r_2)(1 - \cos \theta)}{r_{12}},$$

where θ is the angle between the electron position vectors. The Coulomb divergences cancel, but the local energy depends on triangle shape. Cusp satisfaction is therefore not eigenfunction selection.

For a correction $\psi = \psi_{\text{seed}} e^J$, exactness requires

$$-\frac{1}{2} \sum_{i=1}^2 (\nabla_i^2 J + 2 \nabla_i \ln \psi_{\text{seed}} \cdot \nabla_i J + |\nabla_i J|^2) = E - E_{\text{loc,seed}}.$$

This equation is a starting identity, **not a proposed solution**. The intended breakthrough is a law determining J and E without an unrestricted three-variable solve or newly free coefficients at every order.

Primary test. Derive one previously undetermined global connection relation, predict an unused angular or asymptotic coefficient, and verify it in an independent representation. A better fit to the known energy is not this test.

05

Parallel research objectives

Four independent routes; a common exactness contract.

A / Global connection and resummation

The intended advance. Find a finite generating relation, or a genuinely simpler analytic recursion, that fixes the local expansion's homogeneous data from regularity and decay. Treat logarithmic, angular, and asymptotic sectors together rather than matching only a few leading terms.

First decisive test. Count all still-free functions after each order. Predict a coefficient or connection datum not used to construct the rule. Retire the candidate if every step needs a new unrestricted angular equation with unfixed data.

B / Exact transformation and cancellation

The intended advance. Search coordinate, gauge, or integral transformations that close the kinetic–Coulomb coupling in an explicitly evaluable operator algebra. Retain the transformed measure, boundary conditions, and induced terms; a simpler-looking potential alone is not sufficient.

First decisive test. Prove the map and its inverse on the physical domain. Test the first nontrivial triple-coalescence logarithm, unequal electron radii, and a single-electron escape direction. Any omitted Jacobian or angular term invalidates the candidate.

C / Non-elementary quantization

The intended advance. Seek a special-function, integral, or spectral relation that determines both the energy and the full wavefunction from known kernels and explicitly fixed connection parameters. Do not require elementary radicals as a matter of taste.

First decisive test. Construct the physical branch rule without the benchmark energy. Show how every coefficient is generated and bounded. Reject a disguised full characteristic polynomial or an unresolved interacting resolvent with a new name.

D / An independent continuum minimum proof

The intended advance. Develop a verifier that accepts the constructed function or analytic sequence and checks the full Coulomb problem. Couple symbolic identities with interval bounds where evaluation is numerical; distinguish wavefunction control from an accurate expectation value.

First decisive test. Challenge cusps, collinear limits, triple collision, and remote tails separately. A verifier for a finite matrix is only a finite-matrix verifier until the continuum remainder is proved. Compare with existing atomic lower-bound methods. [\[6\]](#)

Portfolio rule. A candidate survives because it eliminates a defined unknown or proves a genuinely new obstruction. Do not rescue it indefinitely by adding parameters, and do not close its parent idea because one ansatz fails. A stronger hypothesis is welcome only if it supplies a new determining relationship.

06

Prove the state is the minimum

A low residual at sampled points is not a global theorem.

The ideal ground-state certificate

For a positive candidate ψ satisfying $H_2\psi = E\psi$, integration by parts gives, initially for smooth test functions supported away from the collision sets,

$$\langle \varphi, (H_2 - E)\varphi \rangle = \frac{1}{2} \sum_{i=1}^2 \int_{\mathbb{R}^6} \psi^2 \left| \nabla_i \left(\frac{\varphi}{\psi} \right) \right|^2 d^3r_1 d^3r_2 \geq 0.$$

With the domain extension and boundary terms justified, this establishes a lower bound attained by ψ . It selects the ground state without importing an exact excited-state spectrum. The identity is standard ground-state-transform reasoning, not a new result claimed by this charter.

For approximate candidates, a bounded local energy or another rigorous continuum inequality may provide lower and upper bounds. Such bounds are useful, but an open interval does not establish exact closure. [6]

Verification ladder

1. **Algebra and model.** Independently derive the transformed operator and measure. Check every term, parameter convention, and return to the original H_2 .
2. **Singularities and domain.** Check averaged cusps, logarithmic coalescence terms, regularity across coordinate boundaries, kinetic integrability, and the all-space decay conditions.
3. **Parameter determination.** Verify that energy, normalization, and connection data are outputs. Identify and reject spurious or excited branches without benchmark fitting.
4. **Global identity or convergence theorem.** Prove $H_2\psi - E\psi = 0$, or supply explicit global remainders for a convergent analytic construction. A finite precision result must state what it encloses.
5. **Independent numerical stress tests.** Compare energy, density, pair distributions, contact quantities, and residuals in a different coordinate system and implementation. Keep discretization, truncation, quadrature, and rounding errors separate.

Fail-fast controls that do not import the answer

With the electron–electron term removed, the exact normalized spatial state is $(8/\pi)e^{-2(r_1+r_2)}$ and the energy is -4 . Restore the interaction only after reproducing that limit.

For the physical Hamiltonian, an uncorrelated normalized exponential with exponent ζ has expectation

$$E(\zeta) = \zeta^2 - 4\zeta + \frac{5}{8}\zeta.$$

The choice $\zeta = 27/16$ and positivity of electron repulsion give the elementary initial bracket

$$-4 \leq E_0 \leq -\frac{729}{256} < -2.$$

This is a bare-input control, not a precision milestone. Any claimed exact branch inconsistent with it is rejected before a large calculation.

07

Launch plan and decision gates

Organize work around eliminated unknowns, not a count of ideas.

Gate 1 / Establish the exact benchmark contract

Reproduce the operator in two coordinate systems, its measure, cusp limits, and the seed's nonconstant local energy. Distinguish exact identities, convergent expansions, fitted formulas, and claimed global solutions in the literature.

Deliverable: a checked baseline package. Existing energy digits are accessible only to the comparison layer, not to candidate construction.

Gate 2 / Find and attack one new determining relation

Run the tracks independently. Each candidate declares its unknowns and intended elimination. Use exact algebra and local expansions to reject inconsistent proposals before expensive fitting. Numerical discovery is allowed; agreement is not proof.

Promotion test: the relation predicts information not used to obtain it and survives an independent change of geometry, expansion order, or auxiliary coupling within its claimed domain.

Gate 3 / Complete global parameter and branch selection

Connect local structure to physical decay. Determine all parameters without a supplied helium density or energy. Test convergence on the unbounded domain. Use nearby Z or interaction strengths to stress-test the mechanism; solving a whole ion sequence is not a prerequisite.

Promotion test: no unknown function or unrestricted spectral solve remains inside the proposed evaluation rule.

Gate 4 / Assemble the exactness dossier

Deliver the representation, parameter rule, normalization, domain proof, minimum certificate, and independent implementation. Compare the actual contribution with its closest antecedent; an extra identity or energy digit is not automatically a breakthrough.

Closure test: an independent reader can derive and evaluate the result from this dossier without this conversation, private data, or an exact-state oracle.

Record required for every candidate

Identifier and version; domain; inputs and dependencies; claimed elimination; falsification test; proof and numerical status; continuum errors; closest literature; decision and unresolved obligation. Retiring an ansatz does not close its parent question.

Handoff package. Archive derivations, source, inputs, outputs, and an independent verifier under immutable names. Include a manifest and fresh-extraction replay. Label missing evidence; distinguish finite certificates from general proofs.

Do not lower the ambition to save a candidate.

08

Primary literature and use

Selected baselines for this program, not an exhaustive priority audit.

- [1] T. Kato. “On the eigenfunctions of many-particle systems in quantum mechanics.” *Communications on Pure and Applied Mathematics* **10**, 151–177 (1957).
Use: Coulomb eigenfunction regularity and coalescence conditions.
[DOI: 10.1002/cpa.3160100201](https://doi.org/10.1002/cpa.3160100201)
- [2] C. R. Myers, C. J. Umrigar, J. P. Sethna, and J. D. Morgan III. “Fock’s expansion, Kato’s cusp conditions, and the exponential ansatz.” *Physical Review A* **44**, 5537–5546 (1991).
Use: compatibility of cusp-correct exponentials with triple-collision structure.
[DOI: 10.1103/PhysRevA.44.5537](https://doi.org/10.1103/PhysRevA.44.5537)
- [3] C. Schwartz. “Further Computations of the He Atom Ground State.” arXiv:math-ph/0605018 (2006).
Use: high-precision validation reference; not input to an exact construction.
[arXiv: math-ph/0605018](https://arxiv.org/abs/math-ph/0605018)
- [4] A. V. Turbiner, J. C. López Vieyra, J. C. del Valle, and D. J. Nader. “Ultra-Compact accurate wave functions for He-like and Li-like iso-electronic sequences and variational calculus. I. Ground state.” arXiv:2007.11745v4 (2021); journal DOI below.
Use: compact-ansatz baseline and wavefunction-quality comparisons.
[DOI: 10.1002/qua.26586](https://doi.org/10.1002/qua.26586)
- [5] E. Z. Liverts. “Two-electron atomic systems. A simple method for calculating the ground state near the nucleus. Some applications.” arXiv:2410.04900 (2024).
Use: Fock-aware angular coefficients and the remaining eigenproblem.
[arXiv: 2410.04900](https://arxiv.org/abs/2410.04900)
- [6] M. Ronto, P. Jeszenszki, E. Mátyus, and E. Pollak. “Lower bounds on par with upper bounds for few-electron atomic energies.” *Physical Review A* **107**, 012204 (2023).
Use: comparison baseline for rigorous atomic energy verification.
[DOI: 10.1103/PhysRevA.107.012204](https://doi.org/10.1103/PhysRevA.107.012204)
- [7] H. Olivares-Pilón, J. C. López-Vieyra, and A. V. Turbiner. “Helium-like ions (Z, e, e) in Lagrange Mesh method, interpolating the highly-accurate energy spectra vs. Z .” arXiv:2608.10695v1 (11 August 2026), preprint.
Use: recent numerical and fitted-energy benchmarks, not an exact wavefunction.
[arXiv: 2608.10695v1](https://arxiv.org/abs/2608.10695v1) · [preprint](#)

Status at issue. Sources were checked on 12 September 2026. The charter states research goals and acceptance rules; it claims neither a new helium solution nor novelty for the standard identities used to define its tests. The preceding DFT archive motivates the methodology but is not required to understand or execute this program.

Helium: the analytical frontier

A primary-literature audit for the exact-wavefunction research program

Literature cutoff: 12 September 2026. This is a research map, not a new solution, numerical benchmark, or independent proof verification. It contains 34 annotated primary records plus a separate register of reviews and retrieval obligations. The access level is stated for every source.

Central finding. The literature already supplies exact local recurrences, analytically generated global basis families, highly precise energies, and sophisticated energy-bound machinery. The uneliminated step for our target is the constructive global determination of the energy and connection data—not merely the existence of a representation containing them.

The program's ambitious endpoint remains justified as a research target. This audit found no validated compact global construction meeting that endpoint in the material examined. That search result is neither a proof of nonexistence nor a claim that every paper has been examined.

1. Fixed physical target and success standard

The initial target is the normalized, spatially symmetric, positive ground-state amplitude of the nonrelativistic point-nucleus Hamiltonian, in atomic units:

$$H_Z = -\frac{1}{2}(\nabla_1^2 + \nabla_2^2) - Z/r_1 - Z/r_2 + 1/r_{12}, \quad Z = 2.$$

The spin part is the singlet. Finite nuclear mass, relativity and quantum electrodynamics are separate extensions. They cannot be folded into a claim about solving this operator. Total orbital angular momentum zero removes overall orientation, not the dependence on the three internal distances. [R01, R16, R26, R31]

A successful final construction must determine the energy and every global parameter from the stated Hamiltonian. An analytic expression evaluated using a fitted energy, an infinite list of globally fitted coefficients, or an unspecified conditional function does not meet that endpoint. A genuinely evaluated non-elementary construction is admissible. A constructive convergent expansion can be a substantive intermediate advance, but its distinction from existing methods must be proved rather than asserted.

We should distinguish five outcomes: a finite compact approximation; a local analytic expansion; a constructively generated convergent global expansion; an exact integral or differential reformulation with an unresolved solve; and an explicit global determining law. All can be scientifically useful. They do not establish the same claim.

2. Frontier map: what is explicit, and what remains

Approach	What is already constructive	What remains for our endpoint
Perimetric / Hylleraas expansions	Coordinates, analytical basis integrals and coefficient recurrences	Energy selection and a growing global coefficient problem. [R02, R30]
Fock / coalescence analysis	Logarithmic structure and many explicit angular coefficients	Homogeneous terms, global connection, decay and quantization. [R03–R05, R17–R18]
Free Complement	Operator-generated analytic basis; singularity scaling; local-energy and bound checks	Variational or sampled coefficient determination. [R07–R11, R32]
Correlated exponentials / pseudospectral grids	Very precise energies and observable checks	Finite numerical expansion or discretization, not finite exact amplitudes. [R06, R12–R14]
Compact correlated functions	Small formulas informed by cusps and tails	Residual global error and optimized/fitted parameters. [R15–R16, R33–R34]
Two-body Coulomb scattering reduction	Known pair kernels and eliminated frequency integrals	Coupled three-body integral eigenfunctions and energy-root search. R25
Critical-charge / threshold analysis	Accurate thresholds and rigorous asymptotic information	A complete ordinary-helium wavefunction; threshold data alone do not fix it. [R20–R24]

The earliest algebraic reduction must not be overlooked

Pekeris already translated the ground-state problem into a structured perimetric expansion with explicit recurrence relations and a secular condition in 1958. Later code makes this a practical reproducibility baseline. Therefore, obtaining another coefficient recurrence or determinant equation is not automatically a frontier advance. The proposed gain has to concern its solution, convergence, connection data, or complexity. [R02, R30]

3. Coalescence is well understood locally, but does not select the energy

For the helium singlet, electron–nucleus and opposite-spin electron–electron cusps constrain suitable spherical-average derivatives. At simultaneous three-particle coalescence, ordinary power series are insufficient. With $R = \sqrt{r_1^2 + r_2^2}$, the local form contains powers $R^n(\log R)^p$ multiplied by angular functions. A candidate consisting only of a finite analytic polynomial times an ordinary exponential must confront the missing logarithmic structure; this is not an obstruction to general special-function constructions. [R01, R04, R17]

Morgan’s 1986 convergence result is a crucial antidote to a tempting inference. Its abstract states that Fock-type solutions converge pointwise at all radii, but also that infinitely many such PDE solutions exist for arbitrary complex energy when no boundary condition is imposed at infinity. It

explicitly separates the physical decaying-eigenfunction question. We retrieved the abstract and references, not the entire proof or its full subsequent theorem history. The safe conclusion is that convergence of a local representation does not itself quantize the energy. [R03](#)

Recent high-order angular-coefficient calculations provide concrete symbolic material to reuse. Near-nucleus constructions also retain homogeneous constants that must be selected globally, and use generalized eigenvalue calculations. The 2024 Liverts construction is a direct comparator for a cusp/Fock-aware proposal. [R17–R18]

Fournais and collaborators additionally show how an explicit universal factor improves the regularity of the remaining Coulomb wavefunction. That is useful structure, not a formula for the remainder. [R05](#)

4. The major missing antecedent: Free Complement theory

The charter underweighted the scaled Schrödinger equation and Free Complement / free iterative-complement-interaction program. It is not just a numerical basis chosen by trial and error. A scaling factor moderates the singular operator, and repeated operator action generates analytical complement functions. The exact equivalence must be understood with the proper domain and collision behavior. The generated coefficients are then determined variationally or through sampled local Schrödinger equations. [R07, R10, R32]

Nakashima and Nakatsuji's 2008 paper explicitly tests local energy, Hamiltonian variance, and upper/lower energies. It reports 32 agreeing digits in the fixed-nucleus ground energy using a modified Temple construction and an internally computed excited-state estimate. Its four-page paper and relevant figures/tables were inspected. This audit did not independently reproduce the arithmetic or reprove the spectral-label assumptions. [R08](#)

The exponential integral E_i had already been used in 2008 to bring logarithmic and large-distance structure into the generated basis. More recent 2022 and 2024 work investigates improved electron–nucleus/electron–electron scaling, cusp behavior and variational/bound performance. [R09–R11]

Consequence for our claim: “stronger equation → analytically generated functions → cusp tests → energy certificate” is an established architecture. Our contribution would have to determine what it still solves globally, or prove a genuine advantage over it. More terms generated by operator action are not enough.

5. Numerical and compact-form frontiers are different

The idealized helium energy begins $-2.9037243770341195983...$ hartree. Schwartz's 2006 explicitly correlated calculation used 24,099 terms and reported more than forty digits. The Free Complement line supplies an independent high-precision family; correlated exponential calculations in 2018 report 28–35 significant digits across several angular momenta. These are baseline achievements, not an assertion of an all-time accuracy record. [R06–R08, R12]

Pseudospectral calculations check wavefunctions, expectation values and response observables across different regions. They are important because a very good energy does not guarantee equally good contact or tail behavior. A global residual, singular-limit checks, and independent observables should accompany every proposed helium amplitude. Equations satisfied at collocation points are not continuum identities. [R13–R14]

Few-parameter cusp- and tail-aware expressions also have a long history. Kleinekathöfer and collaborators constructed nonvariational boundary-condition-based functions in 1996. Turbiner and collaborators later developed ultra-compact helium-like approximations with millihartree-scale energy accuracy. Forrey’s hybrid Fock/perimetric basis is a particularly relevant prior combination of local and global structure. [R15–R16, R34]

A 2025 generalized-exponential paper measures proximity to the Hartree–Fock limit, not the exact correlated energy. This illustrates why every quoted accuracy needs its reference limit. An August 2026 preprint combines 14–15-digit mesh energies with fitted interpolation across nuclear charge; it supplies neither an exact spatial wavefunction nor a parameter-free quantization rule. [R27, R33]

6. Exact pair physics can be integrated out—but the three-body connection remains

Combescot’s 2017 formulation starts from known two-body Coulomb scattering matrices and derives exact coupled integral equations. This avoids resolving each Coulomb pair from scratch and performs nontrivial analytical integrations. It is a serious precedent for our “eliminate before approximating” strategy. [R25](#)

Its helium implementation still discretizes an integral operator, locates the physical energy through an eigenvalue condition, and reconstructs the solution from the resulting amplitudes. The word “exact” describes the formulation; it does not mean the displayed numerical demonstration is a finite exact wavefunction. Its demonstration should not be compared with forty-digit expansions as though they had the same accuracy target. [R25](#)

A worthwhile attack is therefore specific: identify and eliminate the remaining coupled-kernel dependence, with a domain and physical-root theorem. Renaming the integral eigenfunction or leaving its inverse unevaluated would not advance this step.

7. Global continuation and asymptotic boundary information

The $1/Z$ expansion has long been computed to high order: Baker and collaborators studied 401 orders and the inferred analytic boundary. A finite sequence of accurate coefficients, however, does not independently establish a closed continuation or the exact singularity structure. [R20](#)

The physical binding threshold is approximately $Z_c = 0.911028224$. It must not be conflated with secondary branch parameters proposed on analytically continued sheets. The exact erratum and full later branch analysis need checking before relying on more digits or a rigorous interpretation of fitted singularities. [R21–R24]

A July 2026 publication by Hundertmark, Jex, and Lange gives sharp threshold asymptotic results and threshold-state analysis. It is highly relevant to boundary conditions precisely where ordinary positive-gap decay arguments cease to apply. It does not determine helium’s full wavefunction at $Z = 2$. [R23](#)

The ordinary helium tail depends on ionization information; that information cannot be inserted as a known number into a purported parameter-free construction. Varying Z is a valuable diagnostic of branch selection, but solving all Z is not logically required for a genuine exact $Z = 2$ result.

8. Exactness and lower-bound certification need separate audits

Let $E_u = \langle \psi, H \psi \rangle$ and $\sigma^2 = \langle \psi, (H - E_u)^2 \psi \rangle$ for a normalized admissible trial. If μ is a valid lower bound on the next spectral threshold in the selected symmetry sector and $E_u < \mu$, a Temple-type estimate gives:

$$E_u - \sigma^2/(\mu - E_u) \leq E_0 \leq E_u.$$

The assumptions are part of the result. The gap, eigenvalue ordering, continuum integration and arithmetic cannot be treated as free inputs. A small variance is a weighted residual statement, not a uniform local-energy bound everywhere. Zero variance identifies an eigenstate; physical positivity and sector/minimum reasoning still matter. [\[R08, R19\]](#)

Ronto, Jeszenszki, Mátyus, and Pollak’s 2023 study explicitly discusses a helium multistate application where an admissibility condition fails: an apparently converged candidate lower result is not a valid lower bound. That is a methodological warning, not a refutation of all their bounds. Their condition must be checked before using that branch of the construction. [R19](#)

For our program, exact symbolic verification of a finite recurrence, numerical residual control and a continuum ground-state theorem must remain separate evidence labels.

9. Screen “exact solution” claims by the operator they solve

Search result	Reason it does not yet meet our stated target
Kowol, 2026 revision	Uses an adapted density-convolved potential and additional physical terms; it is not a demonstrated solution of the original point-charge Hamiltonian. This is a scope finding, not a full correctness audit. R29
Konovalov, September 2026	Restricts the two individual angular channels to zero. That is smaller than the full correlated total- $L = 0$ space. The author presents a radial model, not our complete solution. R28
Mesh interpolation, August 2026	Fits an energy formula to numerical data; no exact spatial amplitude follows. R27
Compact generalized exponential, 2025	The benchmark is near-Hartree–Fock accuracy. It must not be quoted as proximity to exact correlated helium. R33

Special harmonium, screened-potential, dimensional, or concentric-sphere solutions can be useful controls, but they must first pass the same Hamiltonian-identification test. We found no reason to replace physical quantum mechanics on the basis of the analytical obstruction being investigated.

10. The revised research target

The broad architecture in our charter needs narrowing, not a lower ambition. Three particular gaps remain plausible targets for original work:

Global connection elimination. Start from the actual Fock homogeneous freedoms and existing resummation results. Produce an explicit rule fixing them from normalizability and physical branch selection. A new numerical matching matrix alone is not the endpoint.

Generator resummation. Start from Free Complement or another analytically defined basis generator. Show that its globally determined coefficients are generated by a smaller law. Count the cost of determining the coefficients and the energy, not only the cost of evaluating a final formula.

Pair-kernel completion. Start from an exact two-body Coulomb reduction and analytically evaluate a decisive remaining contraction or kernel. Preserve the equation's domain, full Coulomb interaction, and selected bound-state branch.

These are research judgments, not claims that a finite closure exists. A strict exclusion of a well-defined candidate may be valuable, but failure of that class is not failure of the helium objective.

Admission gates before a new idea is promoted

1. Reproduce a known local term or recurrence independently; compare it with the original method, not only our earlier derivation.
2. List every undetermined parameter/function and how it will be fixed without numerical helium data.
3. Test a global region not used to construct the candidate: another coalescence, an escape direction, or a changed Z .
4. Derive a continuum minimum or error certificate with its spectral and domain assumptions visible.
5. For a claimed advance, identify the exact operation that the closest antecedent still performs numerically and show how the new construction removes it.

A normalizable positive expression with identically constant local energy for the target Hamiltonian would be a decisive success. A systematically certified expansion can be useful, but should be compared honestly with established high-precision and Free Complement methods.

11. Search coverage and residual uncertainty

This audit used topic searches plus backward and forward citation trails. The main chains were Pekeris/perimetric $\rightarrow$ numerical basis implementations; Kato/Fock/Morgan $\rightarrow$ regularity and modern angular coefficients; scaled equation $\rightarrow$ Free Complement $\rightarrow$ Ei/scaling/bounds; three-body Coulomb T matrices $\rightarrow$ integral formulation; and critical charge $\rightarrow$ threshold asymptotics. "Exact helium solution" searches were checked against the Hamiltonian, parameter inputs and actual computational endpoint.

The search includes papers published or posted through 12 September 2026. Publication and arXiv version dates were used instead of generated date strings inside converted manuscripts. Relevant PDF tables and figures were visually inspected when used. Secondary pages were discovery aids; the scientific register uses original papers or author/publisher records.

The strongest unresolved retrieval obligations are concrete. Morgan's full proof and its subsequent theorem chain were not fully audited; the original 1987 partial-resummation papers remain to be retrieved; Forrey and Tye were available at abstract level; a 2025 conference PDF was too large to retrieve; and the precise critical-charge erratum was located but not fully inspected. The original scaled-equation theorem also needs a domain-level reading before proposing a competing transformation.

This is not a bibliometric census and does not claim to cover every numerical basis, neural trial, collision process, excited resonance, relativistic effect or quantum-electrodynamic correction. Those topics are outside the narrow initial closed-ground-state target or require a separate comparison. No old project theorem, numerical energy, or literature certificate was recomputed in this audit.

Confidence: high that the principal directly relevant architectures and several recent developments have been identified; lower for exhaustive priority on any particular new resummation identity. Absence of a located exact closed solution is a search outcome, not an impossibility theorem.

12. Recommended reading order

Begin with **R08 and R11** to understand how much of our original construction-and-verification plan already exists. Then read **R03, R04 and R18** to separate local singular structure from global selection. Read **R25** as the strongest complementary exact pair-reduction route. Read **R19** before designing a lower certificate. Use **R06, R12–R14, R16 and R34** to define matched numerical and compact-function baselines. Finally, use **R20–R24 and R26–R28** to control analytic continuation, tails and model-scope claims.

The program should seek a new determining step at that boundary—not try to rediscover the existence of an analytical expansion.

Annotated primary-source register

The records state what was inspected, what the source supplies, and the unresolved step relative to this program. “Full text inspected” means the relevant sections were read, not that every proof was independently verified.

R01 • T. Kato • 1957

On the eigenfunctions of many-particle systems in quantum mechanics. Communications on Pure and Applied Mathematics 10, 151–177.

Identifier: 10.1002/cpa.3160100201. [Primary record / author copy](#).

Access: Publisher record; theorem used through later primary treatments.

What it supplies: Foundational regularity and coalescence conditions. Interpret cusp derivatives with their spherical-average qualifications.

What remains: Cusps are boundary constraints, not a global construction or quantization rule.

R02 • C. L. Pekeris • 1958

Ground State of Two-Electron Atoms. Physical Review 112, 1649.

Identifier: 10.1103/PhysRev.112.1649. [Primary record / author copy](#).

Access: Publisher abstract inspected.

What it supplies: Perimetric-coordinate orthogonal expansion with explicit coefficient recurrence and an energy determinant. An early constructive replacement of the differential problem by structured algebra.

What remains: The coefficient recurrence still requires the energy and eigenvector determination; a recurrence alone is not a new closed solution.

R03 • J. D. Morgan III • 1986

Convergence properties of Fock’s expansion for S-state eigenfunctions of the helium atom. Theoretica Chimica Acta 69, 181–223.

Identifier: 10.1007/BF00526420. [Primary record / author copy](#).

Access: Publisher abstract and reference list inspected; full proof not retrieved.

What it supplies: Proves pointwise convergence for Fock-type solutions and explicitly distinguishes this from imposing the boundary condition at infinity. The abstract notes infinitely many such PDE solutions for arbitrary complex energy.

What remains: We have not completed a later-theorem audit of every physical-eigenfunction representability question mentioned in 1986. Do not turn this into a claim that convergence fixes the energy.

R04 • C. R. Myers; C. J. Umrigar; J. P. Sethna; J. D. Morgan III • 1991

Fock's expansion, Kato's cusp conditions, and the exponential ansatz. Physical Review A 44, 5537.

Identifier: 10.1103/PhysRevA.44.5537. [Primary record](#) / [author copy](#).

Access: Publisher abstract inspected.

What it supplies: Examines the second-order Fock structure and develops an exponential representation preserving pair cusps at finite truncation.

What remains: Exponentiating cusp/Fock information is an antecedent, not a new mechanism. The global coefficient determination remains.

R05 • S. Fournais; M. Hoffmann-Ostenhof; T. Hoffmann-Ostenhof; T. Ø. Sørensen • 2005

Sharp Regularity Results for Coulombic Many-Electron Wave Functions. Communications in Mathematical Physics 255, 183–227.

Identifier: 10.1007/s00220-004-1257-6; arXiv:math-ph/0312060. [Primary record](#) / [author copy](#).

Access: Primary abstract inspected.

What it supplies: An explicit universal factor separates cusp/logarithmic structure from a remainder with controlled second derivatives.

What remains: Improved regularity does not determine the remainder, nor does it establish that the remainder is globally elementary or analytic.

R06 • C. Schwartz • 2006

Further Computations of the He Atom Ground State. arXiv preprint.

Identifier: arXiv:math-ph/0605018. [Primary record](#) / [author copy](#).

Access: Primary abstract and computational discussion inspected.

What it supplies: A 24,099-term explicitly correlated calculation reports more than forty digits in the idealized helium ground energy.

What remains: Numerical variational expansion, not a finite closed wavefunction. No all-time record claim is made by this audit.

R07 • H. Nakashima; H. Nakatsuji • 2007

Solving the Schrödinger equation for helium atom and its isoelectronic ions with the free iterative complement interaction (ICI) method. Journal of Chemical Physics 127, 224104.

Identifier: 10.1063/1.2801981. [Primary record](#) / [author copy](#).

Access: Author-hosted full text; relevant method and accuracy sections inspected.

What it supplies: The scaled equation generates complement functions; variational coefficient determination produces exceptionally precise helium energies and wavefunctions.

What remains: The remaining linear coefficients and energy are obtained through an expanding numerical problem. Read before proposing operator-generated analytic bases.

R08 · H. Nakashima; H. Nakatsuji · 2008

How Accurately Does the Free Complement Wave Function of a Helium Atom Satisfy the Schrödinger Equation? Physical Review Letters 101, 240406.

Identifier: 10.1103/PhysRevLett.101.240406. [Primary record / author copy](#).

Access: Full four-page paper inspected; table/figure pages visually checked.

What it supplies: Studies local energy and Hamiltonian variance and reports modified-Temple lower and variational upper bounds agreeing to 32 digits. The excited-state lower estimate is generated in the same calculation.

What remains: Published numerical bound construction, not independently rerun here with interval arithmetic. State identification and spectral-gap assumptions must be retained.

R09 · Y. I. Kurokawa; H. Nakashima; H. Nakatsuji · 2008

Solving the Schrödinger equation of helium and its isoelectronic ions with the exponential integral (Ei) function in the free iterative complement interaction method. Physical Chemistry Chemical Physics 10, 4486–4494.

Identifier: 10.1039/B806979B. [Primary record / author copy](#).

Access: Author-hosted full text; relevant construction inspected.

What it supplies: Uses the exponential integral to combine useful logarithmic and asymptotic behavior in a complement expansion.

What remains: Introducing a special function that contains both a logarithm and decay is already explored; the coefficient-selection problem remains numerical.

R10 · H. Nakatsuji; H. Nakashima; Y. I. Kurokawa · 2022

Accurate scaling functions of the scaled Schrödinger equation. Journal of Chemical Physics 156, 014113.

Identifier: 10.1063/5.0077495. [Primary record / author copy](#).

Access: Author-hosted full text; abstract and scaling construction inspected.

What it supplies: Proposes scaling functions with controlled coalescence and large-distance behavior for free-complement generation.

What remains: The scaling function improves the equation/basis but does not by itself supply the exact finite helium wavefunction.

R11 • H. Nakatsuji; H. Nakashima • 2024

Accurate Scaling Functions of the Scaled Schrödinger Equation. II. Variational Examination of the Correct Scaling Functions with the Free Complement Theory Applied to the Helium Atom. Journal of Chemical Theory and Computation 20, 3749–3765.

Identifier: 10.1021/acs.jctc.4c00263. [Primary record](#) / [author copy](#).

Access: Author-hosted full text; method, bounds, and selected tables inspected.

What it supplies: Examines electron–nucleus and electron–electron scaling, variational accuracy, cusp behavior, residuals and lower bounds. Directly overlaps our proposed stronger-formulation/certificate architecture.

What remains: Still solves for expansion coefficients. Its lower-bound comparisons also require careful attention to excited-state inputs and the bound’s applicability conditions.

R12 • D. T. Aznabaev; A. K. Bekbaev; V. I. Korobov • 2018

Nonrelativistic energy levels of helium atoms. Physical Review A 98, 012510.

Identifier: 10.1103/PhysRevA.98.012510. [Primary record](#) / [author copy](#).

Access: Publisher abstract inspected.

What it supplies: Correlated exponential variational calculations report 28–35 significant digits across S, P, D, and F levels.

What remains: Independent numerical benchmark family; precise energies are not an analytic coefficient-elimination rule.

R13 • P. E. Grabowski; D. F. Chernoff • 2010

Pseudospectral Calculation of the Wavefunction of Helium and the Negative Hydrogen Ion. Physical Review A 81, 032508.

Identifier: 10.1103/PhysRevA.81.032508; arXiv:0909.1791. [Primary record](#) / [author copy](#).

Access: Primary full text; method and convergence inspected.

What it supplies: Multi-domain pseudospectral treatment checks energies, wavefunctions and local convergence without a variational energy-only objective.

What remains: Finite collocation/discretization remains; pointwise equations at grid nodes are not a proof of exactness throughout the continuum.

R14 • P. E. Grabowski; D. F. Chernoff • 2011

Pseudospectral Calculation of Helium Wave Functions, Expectation Values, and Oscillator Strength. Physical Review A 84, 042505.

Identifier: 10.1103/PhysRevA.84.042505; arXiv:1107.2150. [Primary record](#) / [author copy](#).

Access: Primary full text; observable/convergence discussion inspected.

What it supplies: Tests quantities probing different regions of the wavefunction, not merely its variational energy.

What remains: Supplies validation targets. A proposed analytic expression must be tested in singular and tail regions as well as in its energy.

R15 • U. Kleinekathöfer; S. H. Patil; K. T. Tang; J. P. Toennies • 1996

Boundary-condition-determined wave function for the ground state of helium and isoelectronic ions. Physical Review A 54, 2840.

Identifier: 10.1103/PhysRevA.54.2840. [Primary record](#) / [author copy](#).

Access: Publisher abstract inspected.

What it supplies: Constructs compact functions from coalescence and asymptotic conditions, with additional correction strategies.

What remains: Matching physical endpoints as a route to a compact approximation is established. Endpoint agreement does not establish the global solution.

R16 • A. V. Turbiner; J. C. López Vieyra; J. C. del Valle; D. J. Nader • 2020

Ultra-Compact accurate wave functions for He-like and Li-like iso-electronic sequences and variational calculus. I. Ground state. International Journal of Quantum Chemistry; arXiv version.

Identifier: 10.1002/qua.26586; arXiv:2007.11745. [Primary record](#) / [author copy](#).

Access: Primary full text; helium ansätze and accuracy discussion inspected.

What it supplies: Few-parameter correlated expressions obtain high accuracy relative to their compactness and investigate cusp/other observables.

What remains: This is the compact-approximation comparator, not an exact finite solution. A new short ansatz requires a matched complexity–accuracy comparison.

R17 • E. Z. Liverts; R. Krivec • 2022

Fock Expansion for Two-Electron Atoms: High-Order Angular Coefficients. Atoms 10, 135.

Identifier: 10.3390/atoms10040135; arXiv:2209.09053. [Primary record](#) / [author copy](#).

Access: Primary full text; scope of angular-coefficient construction inspected.

What it supplies: Derives explicit higher-order leading-log angular coefficients and machinery for further terms.

What remains: Leading-log terms are not the whole hierarchy; remaining angular components and global constants must not be silently omitted.

R18 • E. Z. Liverts • 2024

Two-electron atomic systems. A simple method for calculating the ground state near the nucleus. Some applications. arXiv preprint (title in repository record).

Identifier: arXiv:2410.04900. [Primary record](#) / [author copy](#).

Access: Primary full text; basis, homogeneous constants, and numerical construction inspected.

What it supplies: Combines cusp-aware exponentials, logarithms and explicit Fock information for accurate near-nucleus calculations.

What remains: A generalized eigenproblem and globally selected coefficients remain. This is a direct comparator for any proposed cusp/Fock-aware generator.

R19 • M. Ronto; P. Jeszenszki; E. Mátyus; E. Pollak • 2023

Lower bounds on par with upper bounds for few-electron atomic energies. Physical Review A 107, 012204.

Identifier: 10.1103/PhysRevA.107.012204; arXiv:2301.02827. [Primary record / author copy](#).

Access: Primary full text; lower-bound theory and helium-specific failure inspected.

What it supplies: Develops few-electron lower bounds with correlated Gaussians and analyzes conditions controlling their validity. Helium provides an explicit warning against naive multistate application.

What remains: Convergence of a displayed bound sequence is not sufficient. Excited-state/interlacing assumptions and numerical errors need independent control.

R20 • J. D. Baker; D. E. Freund; R. N. Hill; J. D. Morgan III • 1990

Radius of convergence and analytic behavior of the $1/Z$ expansion. Physical Review A 41, 1247.

Identifier: 10.1103/PhysRevA.41.1247. [Primary record / author copy](#).

Access: Publisher abstract inspected.

What it supplies: Computes a 401-order expansion and studies its analytic behavior and inferred convergence boundary.

What remains: A long recurrence or extrapolated radius is not a closed wavefunction. Later critical-charge and branch analyses must be compared carefully.

R21 • C. S. Estienne; M. Busuttill; A. Moini; G. W. F. Drake • 2014

Critical Nuclear Charge for Two-Electron Atoms. Physical Review Letters 112, 173001; erratum 113, 039902.

Identifier: 10.1103/PhysRevLett.112.173001; erratum 10.1103/PhysRevLett.113.039902. [Primary record / author copy](#).

Access: Publisher abstract inspected; erratum located, full correction not audited.

What it supplies: High-precision determination of the binding threshold, approximately $Z_c = 0.911028224$.

What remains: Threshold location is an auxiliary benchmark, not the global wavefunction. Do not use every printed digit without checking the erratum.

R22 • P. E. Grabowski; K. Burke • 2015

Quantum critical benchmark for electronic structure theory. Physical Review A 91, 032501; preprint title differs.

Identifier: 10.1103/PhysRevA.91.032501; arXiv:1408.2014. [Primary record / author copy](#).

Access: Primary abstract and author-hosted discussion inspected.

What it supplies: Studies density and electronic structure at the two-electron binding threshold.

What remains: A useful extreme-density/decay test; extrapolating an ordinary helium tail to threshold without analysis is unsafe.

R23 • D. Hundertmark; M. Jex; M. Lange • 2026

Quantum Systems at The Brink: Helium-Type Systems. Archive for Rational Mechanics and Analysis 250, 67; published 13 July 2026.

Identifier: 10.1007/s00205-026-02167-7; arXiv:1908.04883. [Primary record / author copy](#).

Access: Primary full text; statements and asymptotic discussion inspected, not a line-by-line proof audit.

What it supplies: Sharp upper/lower asymptotic estimates and threshold-eigenstate analysis where ordinary gap-separated decay arguments fail.

What remains: Provides rigorous global boundary information, not a closed helium ground wavefunction at $Z = 2$.

R24 • A. V. Turbiner; J. C. López Vieyra; H. Olivares-Pilón • 2016

Three-body quantum Coulomb problem: analytic continuation. Modern Physics Letters A 31, 1650156.

Identifier: arXiv:1506.07403. [Primary record / author copy](#).

Access: Primary abstract and stated continuation claims inspected.

What it supplies: Studies a proposed analytic continuation and distinguishes its secondary branch parameter from the physical binding threshold.

What remains: Its fitted/unphysical-sheet conclusions are not adopted here as rigorous quantization or as verified additional physical states.

R25 • R. Combescot • 2017

The 3-body Coulomb problem. Physical Review X 7, 041035.

Identifier: 10.1103/PhysRevX.7.041035; arXiv:1705.06177. [Primary record / author copy](#).

Access: Primary full text; exact equations and numerical helium implementation inspected.

What it supplies: Uses known two-body Coulomb T matrices to derive coupled integral equations for the three-body bound state. Avoids re-solving each pair and eliminates frequency integrations.

What remains: Helium still requires an integral-operator eigenproblem and an energy search. The wavefunction is reconstructed from the numerical solution, not supplied as a finite closed expression.

R26 • S. Sadhukhan; G. Pestka; R. Podeszwa; H. A. Witek • 2025

Elimination of angular dependency in the quantum three-body problem made easy. arXiv preprint, revised version inspected in 2026.

Identifier: arXiv:2506.23962v3. [Primary record / author copy](#).

Access: Primary full text; reduction and computational endpoint inspected.

What it supplies: Eliminates overall orientation for general masses, total angular momentum, and parity, retaining the terms required by the reduced operator.

What remains: Internal three-distance correlations remain. The helium S-state reduction is not the missing analytical solution.

R27 • H. Olivares-Pilón; J. C. López Vieyra; A. V. Turbiner • 2026

Helium-like ions (Z,e,e) in Lagrange Mesh method, interpolating the highly-accurate energy spectra vs. Z. arXiv preprint, 11 August 2026.

Identifier: arXiv:2608.10695v1. [Primary record](#) / [author copy](#).

Access: Primary full text; numerical method and interpolation inputs inspected.

What it supplies: Reports 14–15 significant-digit numerical Lagrange-mesh energies and compact fitted energy interpolations.

What remains: A fitted energy law is not an exact spatial wavefunction or a noncircular analytic quantization rule.

R28 • D. Konovalov • 2026

Radial Models of Helium and Atomic Hydrogen in Two and Three Dimensions. Few-Body Systems 67, 59; published 9 September 2026.

Identifier: 10.1007/s00601-026-02080-5. [Primary record](#) / [author copy](#).

Access: Primary full text; projection and numerical-method scope inspected.

What it supplies: Unifies radial calculations across dimensions and interactions, explicitly restricting helium to individual zero-angular-momentum channels.

What remains: This is not the full correlated total-L = 0 helium space. Its numerical radial energies do not establish a closed solution to our Hamiltonian.

R29 • F. Kowol • 2026

Analytical solution of the Schrödinger equation for the neutral helium atom in the ground state considering the uncertainty principle, vibrational modes and quantum-electrodynamical effects. arXiv preprint, version 9 (January 2026).

Identifier: arXiv:2406.03020v9. [Primary record](#) / [author copy](#).

Access: Primary PDF; Hamiltonian/potential construction inspected for scope.

What it supplies: Claims an analytic treatment with adapted, density-convolved potentials and additional effects. Included as a claim-screening case, not an accepted solution.

What remains: Model mismatch excludes it as a demonstration for the original point-charge nonrelativistic Hamiltonian. This is not a complete correctness audit of the paper.

R30 • E. Z. Liverts; N. Barnea • 2011

S-states of helium-like ions. Computer Physics Communications 182, 1790–1795; author software deposit.

Identifier: Mendeley Data k7f2hk7fc2/1. [Primary record](#) / [author copy](#).

Access: Author software-deposit abstract inspected; code not executed.

What it supplies: Supplies a Mathematica implementation of the Pekeris/perimetric method with generalized matrix eigenvalues and eigenvectors.

What remains: Useful reproduction baseline, but numerical coefficient determination remains. Package access and executable environment should be checked before a benchmark run.

R31 • G. W. F. Drake; A. T. Bondy; J. A. Hallett; M. Najem • 2026

Theory for the Rydberg states of helium: Results and comparison with precision spectroscopy. Physical Review A; title abbreviated.

Identifier: 10.1103/1dbh-rjnn; arXiv:2510.17495. [Primary record](#) / [author copy](#).

Access: Publisher abstract inspected.

What it supplies: High-precision Hylleraas calculations for Rydberg states demonstrate the broader numerical/spectroscopic frontier, including effects outside the initial idealized Hamiltonian.

What remains: Not a ground-state closed representation. Nuclear, relativistic and QED contributions must remain separate from our first target.

R32 • H. Nakatsuji • 2004

Scaled Schrödinger-equation foundation of the Free Complement approach. Physical Review Letters 93, 030403; descriptive short title.

Identifier: 10.1103/PhysRevLett.93.030403. [Primary record](#) / [author copy](#).

Access: Original reference identified in inspected 2008/2024 primary papers; original full text not audited.

What it supplies: Introduces the scaled-equation route that motivates later Free Complement calculations.

What remains: Historical antecedent requiring original-theorem/domain inspection before proposing a competing scaling transformation.

R33 • J. W. Tye • 2025

Analysis of Compact, Generalized Exponential Basis Functions for Helium. Few-Body Systems 66, 42; published 7 October 2025.

Identifier: 10.1007/s00601-025-02008-5. [Primary record](#) / [author copy](#).

Access: Publisher abstract and reference list inspected; full article not retrieved.

What it supplies: Optimizes generalized exponential atomic functions near the Hartree–Fock limit and examines radial and cusp properties.

What remains: Its stated 0.20 millihartree proximity is to the Hartree–Fock limit, not to the exact correlated helium energy. Do not mistake a compact-orbital result for a full correlated solution.

R34 • R. C. Forrey • 2004

Compact representation of helium wave functions in perimetric and hyperspherical coordinates. Physical Review A 69, 022504.

Identifier: 10.1103/PhysRevA.69.022504. [Primary record](#) / [author copy](#).

Access: Author-institution publication abstract inspected; primary full text not retrieved.

What it supplies: Combines an explicit Fock/hyperspherical component with perimetric Laguerre functions to improve variational convergence.

What remains: Important prior hybrid of local singular structure and global basis. Full details still needed for an implementation-level novelty comparison.

Reviews and outstanding retrieval leads

These are not used as unexamined proof authorities. Follow their primary references or retrieve the missing original before making a detailed novelty claim.

L01 • Atomic structure calculations of helium with correlated exponential functions

V. A. Yerokhin; V. Patkóš; K. Pachucki; 2021. [Record / retrieval target](#).

Author review; abstract inspected; used to route QED/spectroscopy literature, not as an independent original theorem.

L02 • Exact factorization of a many-body wavefunction beyond the electron-nuclear problem

P. Schürger; S. Giarrusso; F. Agostini; 2026. [Record / retrieval target](#).

Review; relevant electronic-factorization sections inspected. Coupled conditional amplitudes/potentials remain. Trace any claim to the original papers before treating it as a new helium method.

L03 • 2025 Warsaw conference: analytical helium/Fock progress

Specific abstract requires retrieval; 2025. [Record / retrieval target](#).

Search snippet refers to Fock's implicit form; the large PDF exceeded the web fetch limit, and container download failed. No detailed result is claimed.

L04 • Partial resummation of the Fock expansion and explicit second-order term

P. C. Abbott; E. N. Maslen; J. E. Gottschalk (two linked 1987 papers); 1987. [Record / retrieval target](#).

Historical references identified in later primary work; original proofs not retrieved. Needed for a detailed resummation-priority audit.

L05 • Full-text follow-up for R33 and R34

J. W. Tye; R. C. Forrey; 2004 / 2025. [Record / retrieval target](#).

Publisher/author abstracts now retrieved and registered. The full basis constructions were not obtained; no numerical implementation or detailed equivalence audit is claimed.